Singular Learning Theory and Deep Neural networks

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Abstract

Singular Learning Theory was developed by Sumio Watanabe(Watanabe, 2009). Using the resolution of singularities, he analyzed the asymptotic expansion of generalization errors, training errors, free energy, and other quantities with respect to the number of data samples. To characterize the leading term, he introduced the concept of learning coefficients and their orders, which play a crucial role in learning theory. In this paper, we first examine the learning coefficients, which correspond to log canonical thresholds in algebraic geometry, for singular learning models, and then show the another proof for Watanabe theory, which is useful for understanding his theory deeply.

1. Introduction

Machine learning is the process of inferring the probability distribution (true distribution) of an information source from a large amount of data. The systematic framework that organizes and formalizes the mechanisms of learning is known as learning theory. Models with hierarchical or internal structures that can represent a wide range of probability distributions by varying parameters are called singular models. These models exhibit complex structures. For instance, learning data used in image and speech recognition, genetic analysis, time-series prediction, and data mining do not typically follow simple distributions such as the normal distribution; instead, they possess highly complex structures. Since these complexities cannot be adequately captured within the framework of classical theories, research on novel theoretical approaches has rapidly expanded. In particular, studies on the learning coefficient, which represents learning efficiency, by using the resolution of singularities theorem have gained increasing importance. This paper primarily explores learning coefficients in Bayesian machine learning. These studies have been developed through the application of algebraic geometry to learning theory. Recently, numerous studies on deep learning have emerged. However, many of them are largely numerical and empirical. Moving forward, theoretical research in this area is expected to play an increasingly vital role from an analytical perspective.

2. Singular learning theory

Let us assume that each sample (x_i, y_i) is drawn from a probability density function $q(x, y)$, and that the training set $(x, y)^n := \{(x_i, y_i)\}_{i=1}^n$ consists of n samples selected independently and identically from $q(x, y)$. To estimate the true probability density function $q(x, y)$ using $(x, y)^n$ within the framework of Bayesian estimation, we consider a learning model expressed in probabilistic form as $p(x, y|w)$, along with a prior probability density function $\varphi(w)$ defined on a compact parameter set W, where $w \in W \subset \mathbb{R}^d$ is a parameter. The *a posteriori* probability density function $p(w|(x, y)^n)$ is then given by:

$$p(w|(x,y)^n) = \frac{1}{Z_n(\beta)}\varphi(w)\prod_{i=1}^{n} p(x_i, y_i|w)^{\beta},$$

where

$$Z_n(\beta) = \int_W \varphi(w)\prod_{i=1}^{n} p(x_i, y_i|w)^{\beta}\, dw,$$

with β representing an inverse temperature.

The Kullback-Leibler divergence,

$$D(p_1 \mid p_2) = \int p_1(z)\log\frac{p_1(z)}{p_2(z)}\, dz,$$

serves as a pseudo-distance between arbitrary probability density functions $p_1(z)$ and $p_2(z)$. The optimal parameter w_0 minimizes

$$L(w) = -E_{x,y}[\log p(x, y|w)]$$

$$= \int q(x,y)\log\frac{q(x,y)}{p(x,y|w)}dxdy - \int q(x,y)\log q(x,y)dxdy$$

$$= D(q(x,y) \mid p(x,y|w)) - \int q(x,y)\log q(x,y)dxdy.$$

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Thus, w_0 minimizes $D(q(x,y) \mid p(x,y|w))$, and we define

$$W_0 = \{w \in W \mid L(w) = \min_{w' \in W} L(w')\}.$$

We assume that the log-likelihood function has relatively finite variance, satisfying

$$E_{x,y}\left[\log \frac{p(x,y|w_0)}{p(x,y|w)}\right] \geq c\, E_{x,y}\left[\left(\log \frac{p(x,y|w_0)}{p(x,y|w)}\right)^2\right],$$

for $w_0 \in W_0$, $w \in W$, and some constant $c > 0$. Under this assumption, there exists a unique probability density function $p_0(x,y) = p(x,y|w_0)$ for all $w_0 \in W_0$. Now, define

$$f(x,y|w) = \log \frac{p_0(x,y)}{p(x,y|w)}$$

and its expected error function

$$K(w) = E_{x,y}[f(x,y|w)].$$

Clearly, $K(w_0) = 0$ for all $w_0 \in W_0$.

In Bayesian estimation, the predictive probability density function of $(x,y)^n$ is given by:

$$p((x,y)^n) = Z_n(1) = \int \prod_{i=1}^n p(x_i, y_i|w)\varphi(w)dw.$$

For $w_0 \in W_0$, we have

$$\begin{aligned} & p((x,y)^n) \\ = & \prod_{i=1}^n p(x_i,y_i|w_0) \int \prod_{i=1}^n \frac{p(x_i,y_i|w)}{p(x_i,y_i|w_0)}\varphi(w)dw \\ = & \prod_{i=1}^n p(x_i,y_i|w_0) \int \exp(-nK_n)\varphi(w)dw, \end{aligned}$$

where we define the empirical process $K_n(w)$ as

$$nK_n(w) = \sum_{i=1}^n \log f(x_i,y_i|w).$$

By applying Hironaka ’ s theorem (Hironaka, 1964) to the function $K(w)$ at w_0, we obtain a proper analytic map π from a manifold Y to a neighborhood of $w_0 \in W_0$:

$$K(\pi(u)) = u_1^{2k_1} u_2^{2k_2} \cdots u_d^{2k_d},$$

where $(u_1, \ldots, u_d)$ is a local analytic coordinate system on $U \subset Y$. Furthermore, there exist analytic functions $a(x,y|u)$ and $b(u) \neq 0$ such that

$$f(x,y|\pi(u)) = u_1^{k_1} u_2^{k_2} \cdots u_d^{k_d} a(x,y|u),$$

and

$$\pi'(u)\varphi(\pi(u)) = u_1^{h_1} u_2^{h_2} \cdots u_d^{h_d} b(u).$$

Now, let

$$\xi_n(u) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \left\{ u_1^{k_1} u_2^{k_2} \cdots u_d^{k_d} - a(x_i, y_i|\pi(u)) \right\}.$$

Then, we obtain

$$nK_n(\pi(u)) = \sum_{i=1}^n f(x_i,y_i|\pi(u))$$

$$= nu_1^{2k_1} u_2^{2k_2} \cdots u_d^{2k_d} - \sqrt{n} u_1^{k_1} u_2^{k_2} \cdots u_d^{k_d} \xi_n(u).$$

$$= nK(\pi(u)) - \sqrt{n}\sqrt{K(\pi(u))}\xi_n(u).$$

Definition 2.1. We introduce learning coefficients

$$\lambda(w_0) = \min_u \min_{1 \leq j \leq d} \frac{h_j + 1}{2k_j},$$

and its order

$$\theta(w_0) = \max_u \mathrm{Card}(\{j : \frac{h_j+1}{2k_j} = \lambda\}),$$

where $\mathrm{Card}(S)$ denotes the cardinality of a set S.

Theorem 2.2. *(Watanabe, 2009)*

We obtain

$$\begin{aligned} & \int \exp(-nK_n)\varphi(w)\,dw = \frac{(\log n)^{\theta(w_0)-1}}{n^{\lambda(w_0)}} \\ & \times \int_0^\infty dt \quad t^{\lambda(w_0)-1} \exp(-t + \sqrt{t}\xi_n(u))\,du^* \\ & + o_p(\frac{(\log n)^{\theta(w_0)-1}}{n^{\lambda(w_0)}}). \end{aligned}$$

where $\mu_j = -2\lambda k_j + h_j$,

$$du^* = \frac{\prod_{i=1}^\theta \delta(u_i) \prod_{j=\theta+1}^d u_j^{\mu_j}}{(\theta-1)! \prod_{i=1}^\theta (2k_i)} b(u)du,$$

and $\delta(u)$ *is Dirac's delta function.*

Therefore, we have

$$\begin{aligned} F_n &= -\log p((x,y)^n) \\ &= -\log(\prod_{i=1}^n p(x_i,y_i|w_0)) - \log \int \exp(-nK_n)\varphi(w)dw \\ &= -\log(\prod_{i=1}^n p(x_i,y_i|w_0)) \\ &\quad + \lambda(w_0)\log(n) - (\theta(w_0)-1)\log\log(n) + o_p(1) \end{aligned}$$

which is the free energy.

From these relations, the Kullback-Leibler divergence between $q((x,y)^n) = \prod_{i=1}^n q(x_i, y_i)$ and $p((x,y)^n)$ can be expressed in terms of $\lambda(w_0)$ and $\theta(w_0)$ as follows:

$$\begin{aligned}
& D(q((x,y)^n) \mid p((x,y)^n)) \\
= & \int q((x,y)^n) \log \frac{q((x,y)^n)}{p((x,y)^n)} \prod_{i=1}^n dx_i dy_i \\
= & \int q((x,y)^n) \log q((x,y)^n) \prod_{i=1}^n dx_i dy_i \\
& - \int q((x,y)^n) \log \prod_{i=1}^n p(x_i, y_i | w_0) \prod_{i=1}^n dx_i dy_i \\
& + \int q((x,y)^n) \log \frac{\prod_{i=1}^n p(x_i, y_i|w_0)}{p((x,y)^n)} \prod_{i=1}^n dx_i dy_i \\
= & \int q((x,y)^n) \log q((x,y)^n) \prod_{i=1}^n dx_i dy_i \\
& - \int q((x,y)^n) \log \prod_{i=1}^n p_0(x_i, y_i) \prod_{i=1}^n dx_i dy_i \\
& + \lambda(w_0) \log(n) - (\theta(w_0) - 1) \log\log(n) + O_p(1).
\end{aligned}$$

The Kullback-Leibler divergence serves as a measure of the quality of the learning model $p(x|w)$ and the prior distribution $\varphi(w)$. These results lead to the derivation of the information criteria known as the "widely applicable Bayesian information criterion" (WBIC) (Watanabe, 2013) and the "singular Bayesian information criterion" (sBIC) (Drton, 2012).

Our aim in this paper is to examine λ and θ for deep-layered neural networks (Aoyagi, 2024) with linear units and the ReLU (Rectified Linear Unit) activation function.

Recently, these values have been utilized in (Furman & Lau, 2024), which empirically developed a method for obtaining the local learning coefficient in deep linear networks. Exact or bounded values of λ and θ for Vandermonde matrix-type singularities have been derived in (Aoyagi & Watanabe, 2005a; Aoyagi, 2006; 2013b; 2019b;a). Furthermore, these coefficients have been studied in (Aoyagi, 2013a; Rusakov & Geiger, 2002; 2005; Zwiernik, 2011; Drton et al., 2017), each addressing different aspects. However, these results apply to three-layer neural networks, normal mixture models, and other structures rather than deep linear networks. Therefore, this study is the first to obtain λ and θ for deep-layered neural networks with linear and ReLU units.

3. Deep neural network with linear units and ReLU units

The learning coefficients correspond to log canonical thresholds in algebraic geometry. These thresholds were initially studied theoretically over the complex field or algebraically closed fields in mathematics, where their values can be determined using Hironaka's theorem (Kollár, 1997; Mustata, 2002; Kashiwara, 1976; Fulton, 1993). However, learning coefficients correspond to log canonical thresholds in the real field, making their determination a challenging and intriguing problem, even from a mathematical perspective.

Define matrices $A^{(s)}$ of size $H^{(s)} \times H^{(s+1)}$ and vectors $B^{(s)}$ of dimension $H^{(s)}$, for $s = 1, \ldots, L$:

$$\begin{aligned}
A^{(s)} &= (a_{ij}^{(s)}), \qquad (1 \le i \le H^{(s)}, 1 \le j \le H^{(s+1)}), \\
B^{(s)} &= (b_j^{(s)}), \qquad (1 \le i \le H^{(s)}).
\end{aligned}$$

Now, consider a deep neural network with linear units, where there are $H^{(1)}$ input units, $H^{(L+1)}$ output units, and $H^{(s)}$ hidden units in each hidden layer. We define

$$F^{(s)}(x) = A^{(s)}x + B^{(s)}$$

and consider an input $x \in \mathbb{R}^{H^{(L+1)}}$ with probability density function $q(x)$, and an output $y \in \mathbb{R}^{H^{(1)}}$, where

$$\begin{aligned}
h(x, A, B) &= F^{(1)} \circ F^{(2)} \circ \cdots \circ F^{(L)}(x) \\
&= \prod_{s=1}^{L} A^{(s)} x + \sum_{S=2}^{L} \prod_{s=1}^{S-1} A^{(s)} B^{(S)} + B^{(1)}.
\end{aligned}$$

The learning model with Gaussian noise is given by

$$p(x, y|w) = q(x) \frac{1}{(\sqrt{2\pi})^{H^{(1)}}} \exp(-\frac{1}{2}||y - h(x, A, B)||^2).$$

Assume that one of the optimal parameters is $w^* = (A^{*(s)}, B^{*(s)})$, where the rank of $\prod_{s=1}^{L} A^{*(s)}$ is r, and that the $C^\infty-$ *a priori* probability density function $\varphi(w)$ satisfies $\varphi(w^*) > 0$.

Then, the log canonical threshold λ and its order θ are given as follows.

Theorem 3.1. *Define $M^{(s)} = H^{(s)} - r$ for $s = 1, \ldots, L+1$. Let $\mathcal{M} = \{S_1, \ldots, S_{\ell+1}\} \subset \{1, \ldots, L+1\}$ with $\ell = \mathrm{Card}(\mathcal{M}) - 1$ such that*

$$\sum_{k=1}^{\ell+1} M^{(S_k)} \ge \ell M^{(s)} \text{ for } s \in \mathcal{M},$$
$$\sum_{k=1}^{\ell+1} M^{(S_k)} < \ell M^{(s)} \text{ for } s \notin \mathcal{M}.$$

and let $a = \sum_{k=1}^{\ell+1} M^{(S_k)} - \ell(\lceil \frac{\sum_{k=1}^{\ell+1} M^{(S_k)}}{\ell} \rceil - 1)$.

Then we have

$$\lambda = \frac{H^{(1)}}{2} + \lambda(H^{(s)}, r)$$

and $\theta = \theta(H^{(s)}, r)$, *where*

$$\lambda(H^{(s)}, r) = \frac{-r^2 + r(H^{(1)} + H^{(L+1)})}{2} + \frac{a(\ell - a)}{4\ell} + \frac{1}{4\ell}(\sum_{j=1}^{\ell+1} M^{(S_j)})^2 - \frac{1}{4}\sum_{j=1}^{\ell+1}(M^{(S_j)})^2$$

and $\theta(H^{(s)}, r) = a(\ell - a) + 1$.

The above theorem is derived from the Main Theorem in (Aoyagi, 2024).

Next, we consider the case in multiple-layered neural networks with ReLU units.

For a matrix $A = (a_{ij}), a_{ij} \in \mathbf{R}$, define

$$A_+ = (\max\{0, a_{ij}\}).$$

Definition 3.2. Let X be the region of the input x, and define

$$F^{*(s)}(x) = A^{*(s)}x + B^{*(s)}.$$

Also, define a subset of the region of x as

$$V^{(L)}(F^{*(L)}_{i_1}, \ldots, F^{*(L)}_{i_k}) = \left\{ x \,\middle|\, \begin{array}{l} F^{*(L)}_{i_1}(x) \geq 0, \ldots, F^{*(L)}_{i_k}(x) \geq 0, \\ F^{*(L)}_{j}(x) < 0 \text{ for } j \notin \{i_1, \ldots, i_k\} \end{array} \right\}.$$

The set of such regions is defined as

$$\Omega^{(L)} = \Big\{ V \,\Big|\, V = V^{(L)}(F^{*(L)}_{i_1}, \ldots, F^{*(L)}_{i_k}) \text{ for some } 1 \leq i_1, i_2, \ldots, i_k \leq H^{(L)} \Big\}.$$

Inductively, we define

$$V^{(s)}(F^{(s)}_{i_1}, \ldots, F^{(s)}_{i_k}) = \Big\{ x \in V^{(s+1)} \in \Omega^{(s+1)} \,\Big|\, \begin{array}{l} F^{(s)}_{i_1} \circ F^{(s+1)} \circ \cdots \circ F^{(L)}(x) \geq 0, \\ \vdots \\ F^{(s)}_{i_k} \circ F^{(s+1)} \circ \cdots \circ F^{(L)}(x) \geq 0, \\ F^{(s)}_{j} \circ F^{(s+1)} \circ \cdots \circ F^{(L)}(x) < 0 \text{ for } j \notin \{i_1, \ldots, i_k\} \end{array} \Big\}.$$

Similarly,

$$\Omega^{(s)} = \Big\{ V \,\Big|\, V = V^{(s)}(F^{(s)}_{i_1}, \ldots, F^{(s)}_{i_k}) \text{ for some } 1 \leq i_1, i_2, \ldots, i_k \leq H^{(s)} \Big\}.$$

By the definition, we have $V_1 \cap V_2 = \emptyset$ for $V_1, V_2 \in \Omega^{(s)}$.

Let $h_+(x, A, B) = F^{(1)}_+ \circ F^{(2)}_+ \circ \cdots \circ F^{(L)}_+(x)$.

Theorem 3.3. *If for all* $x \in X$,

$$F^{(s)}_{i+} \circ \cdots \circ F^{(L)}_{+}(x) = 0$$

in a neighborhood of

$$\{A^{*(s)}, A^{*(s+1)}, \ldots, A^{*(L)}, B^{*(s)}, B^{*(s+1)}, \ldots B^{*(L)}\},$$

then, we can remove the i*th row of* $A^{(s)}$*, the* i*th element of* $B^{(s)}$*, and the* i*th column of* $A^{(s-1)}$*. This removal corresponds to eliminating the* i*th unit in the* s*th hidden layer.*

We can rewrite $K(w)$ using $\Omega^{(1)}$ as follows:

$$K(w) = \int_X ||h_+(x, A, B) - h_+(x, A^*, B^*)||^2 q(x)dx$$

$$= \sum_{V \in \Omega^{(1)}, |V|>0} \int_V ||h_+(x, A, B) - h_+(x, A^*, B^*)||^2 q(x)dx.$$

Example 3.4. *Let* $A^{*(1)} = \begin{pmatrix} 1 & 1 & 1 & 1 \end{pmatrix}$, $A^{*(2)} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & -1 \\ -1 & -1 \end{pmatrix}$ *and* $B^{*(2)} = \begin{pmatrix} -2 \\ -5 \\ 1 \\ -1 \end{pmatrix}$. *Also let* $x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$, $x_1, x_2 \geq 0$.

Consider the case of

$$y = A^{*(1)}(A^{*(2)}x + B^{*(2)})_+.$$

We assume that x_1 *and* x_2 *are positive, and therefore* $-x_1 - x_2$ *is never positive. So the the* 4*th row of* $A^{(2)}$*, the* 4*th element of* $B^{(2)}$*, and the* 4*th column of* $A^{(1)}$ *can be removed.*

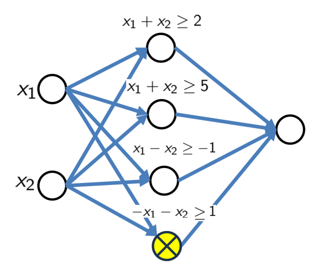

Figure 1. Example 3.4 : 4the units in hidden layer can be removed.

Then the elements of $\Omega^{(2)}$ *are as follows:*

$$\begin{aligned}
V^{(2)}_1 &= \{x_1 + x_2 \geq 2, x_1 + x_2 \geq 5, x_1 - x_2 \geq -1\}, \\
V^{(2)}_2 &= \{x_1 + x_2 \geq 2, x_1 + x_2 < 5, x_1 - x_2 \geq -1\}, \\
V^{(2)}_3 &= \{x_1 + x_2 \geq 2, x_1 + x_2 \geq 5, x_1 - x_2 < -1\}, \\
V^{(2)}_4 &= \{x_1 + x_2 \geq 2, x_1 + x_2 < 5, x_1 - x_2 < -1\}, \\
V^{(2)}_5 &= \{x_1 + x_2 < 2, x_1 + x_2 < 5, x_1 - x_2 \geq -1\}, \\
V^{(2)}_6 &= \{x_1 + x_2 < 2, x_1 + x_2 < 5, x_1 - x_2 < -1\}.
\end{aligned}$$

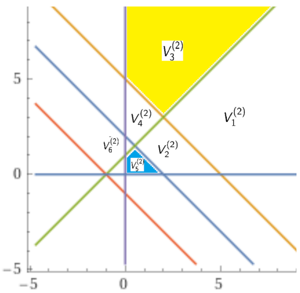

Figure 2. Example 3.4 : The set of divided regions in $\Omega^{(2)}$.

Therefore we have

$$K(w) = \int_X ||h_+(x,A,B) - h_+(x,A^*,B^*)||^2 q(x)dx$$

$$= \int_{V_1^{(2)}\cup\cdots\cup V_6^{(2)}} ||h_+(x,A,B) - h_+(x,A^*,B^*)||^2 q(x)dx$$

$$\cong \left\| A^{(1)} \begin{pmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \\ 0 & 0 & 0 \end{pmatrix} - A^{*(1)} \begin{pmatrix} 1 & 1 & -2 \\ 1 & 1 & -5 \\ 1 & -1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \right\|^2_{V_1^{(2)}}$$

$$+ \left\| A^{(1)} \begin{pmatrix} a_{11} & a_{12} & b_1 \\ 0 & 0 & 0 \\ a_{31} & a_{32} & b_3 \\ 0 & 0 & 0 \end{pmatrix} - A^{*(1)} \begin{pmatrix} 1 & 1 & -2 \\ 0 & 0 & 0 \\ 1 & -1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \right\|^2_{V_2^{(2)}}$$

$$+ \left\| A^{(1)} \begin{pmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - A^{*(1)} \begin{pmatrix} 1 & 1 & -2 \\ 1 & 1 & -5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right\|^2_{V_3^{(2)}}$$

$$+ \left\| A^{(1)} \begin{pmatrix} a_{11} & a_{12} & b_1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - A^{*(1)} \begin{pmatrix} 1 & 1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right\|^2_{V_4^{(2)}}$$

$$+ \left\| A^{(1)} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ a_{31} & a_{32} & b_3 \\ 0 & 0 & 0 \end{pmatrix} - A^{*(1)} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & -1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \right\|^2_{V_5^{(2)}}.$$

Using the generation of ideal, we have

$$K(w) \cong \left\| a_2^{(1)} \begin{pmatrix} a_{21} & a_{22} & b_2 \end{pmatrix} - a^{*(1)}_2 \begin{pmatrix} 1 & 1 & -5 \end{pmatrix} \right\|^2$$

$$+ \left\| a_1^{(1)} \begin{pmatrix} a_{11} & a_{12} & b_1 \end{pmatrix} - a^{*(1)}_1 \begin{pmatrix} 1 & 1 & -2 \end{pmatrix} \right\|$$

$$+ \left\| a_3^{(1)} \begin{pmatrix} a_{31} & a_{32} & b_3 \end{pmatrix} - a^{*(1)}_3 \begin{pmatrix} 1 & -1 & 1 \end{pmatrix} \right\|^2,$$

and we then have $\lambda = \frac{9}{2}$, $\theta = 1$.

By using ReLU units, the input region X is partitioned, and learning is efficiently performed with linear units in each subset. In other words, the learning coefficient can be obtained as the sum of the learning coefficients of the linear units. So we have the following theorem for the case in three-layered neural networks with activation function ReLU.

Theorem 3.5. *Let*

$$h_+(x,A,B) \quad = \quad A^{(1)}(A^{(2)}x + B^{(2)})_+$$

Let us divide the hidden layer into m groups:

$$H_1^{(2)} + H_2^{(2)} + \cdots + H_m^{(2)} = H'^{(2)}.$$

Let $H'^{(1)}_i$ and $H'^{(2)}$ be the number such that $H^{(1)} - H'^{(1)}_i$ and $H^{(2)} - H'^{(2)}$ represent the number of removed neurons which satisfy the condition in Theorem 3.3 for $1 \le i \le m$, respectively.

Then, the learning coefficient λ, which corresponds to the log canonical threshold, is given by

$$\lambda = \sum_{i=1}^m \lambda(H'^{(1)}_i, H_i^{(2)}, H^{(3)} + 1, r_i),$$

and

$$\theta = \sum_{i=1}^m (\theta(H'^{(1)}_i, H_i^{(2)}, H^{(3)} + 1, r_i) - 1) + 1,$$

where r_i is the rank corresponding to group i.

This is the first result obtained for the case where the activation function is nonlinear.

4. δ function in learning theory

In this section, we present a new proof for the δ function in learning theory

Definition 4.1. Let $U \subset \mathbf{R}^d$. For a C^1 function F_1 and a function φ defined on U, we define the level set

$$U_t = \{u \in U \mid F_1(u) = t\}.$$

Let F_2 be a function mapping u to $v \in \mathbb{R}^{d-1}$, and consider the change of variables

$$F(u) = (F_1(u), F_2(u)) = (t, v).$$

Then, the integral transformation is defined as

$$\int_U \delta(t - F_1(u))\varphi(u)\,du = \int_{U_t} \varphi(t,v) \frac{1}{\left|\frac{\partial F}{\partial u}\right|} dv.$$

Here, $\left|\frac{\partial F}{\partial u}\right|$ denotes the Jacobian determinant.

By the definition,

$$\int_U \delta(t - F_1(u))\varphi(u)\, du$$

can be regarded as the integral over variables other than t, such that

$$\int \left(\int_U \delta(t - F_1(u))\varphi(u)\, du \right) dt = \int \varphi(u)\, du.$$

Now, we consider the integral

$$I = \int \exp(-nK(w))\varphi(w)\, dw.$$

Using the analytic transformation $w = \pi(u)$ from Hironaka ' s theorem, we rewrite the integral in terms of the new local coordinates u:

$$I = \sum_u \int \exp\left(-nu_1^{2k_1} \cdots u_d^{2k_d}\right) u_1^{h_1} \cdots u_d^{h_d} b(u)\, du.$$

To analyze I, we introduce the transformation $t' = nK(\pi(u)) = nu_1^{2k_1} \cdots u_d^{2k_d}$ and evaluate its order in terms of n. First, we perform the variable change $t = K(\pi(u)) = u_1^{2k_1} \cdots u_d^{2k_d}$, followed by the transformation $t' = nt$. The proof presented in (Watanabe, 2009), which elegantly employs Mellin transforms and zeta functions, can actually be proven quite easily and understood, even by undergraduate students.

Theorem 4.2. *Let* $U = [0, 1]^d$*, and define* $t = F_1(u) = u_1^{2k_1} \cdots u_d^{2k_d}$ *and* $F_2(u) = v = (u_2, \ldots, u_d)$*.*

Here, assume that $k_1 \neq 0, \ldots, k_{d'} \neq 0$ *and* $k_{d'+1} = \cdots = k_d = 0$*, and let* $\varphi(u) = u^h$*. Additionally, assume that*

$$\lambda = \frac{h_1 + 1}{2k_1} = \cdots = \frac{h_\theta + 1}{2k_\theta}$$

$$< \frac{h_{\theta+1} + 1}{2k_{\theta+1}} \leq \frac{h_{\theta+2} + 1}{2k_{\theta+2}} \leq \cdots \leq \frac{h_{d'} + 1}{2k_{d'}}.$$

Define

$$\mu_{\min} = -2\lambda + \frac{h_{\theta+1} + 1}{k_{\theta+1}}, \quad \mu_{\max} = -2\lambda + \frac{h_{d'} + 1}{k_{d'}}.$$

Then, let $\mathrm{Main}(t) =$

$$\frac{t^{\lambda-1} \log^{\theta-1}(t^{-1})}{(\theta - 1)! 2^\theta \prod_{j=1}^{d'} k_j \prod_{j=\theta+1}^{d'} (-2\lambda + \frac{h_j+1}{k_j}) \prod_{j=d'+1}^{d} (h_j + 1)}.$$

As $t \to 0$*, we obtain*

$$\int_U \delta(t - u^{2k})\varphi(u)\, du = \mathrm{Main}(t)$$

$$+ \begin{cases} O(t^{\lambda-1+\mu_{\min}}) + O(t^{\lambda-1} \log^{\theta-2}(t^{-1})), & \theta \geq 2, \\ O(t^{\lambda-1+\mu_{\min}}), & \theta = 1. \end{cases}$$

Furthermore, the difference between $\int_U \delta(t - u^{2k})\varphi(u)\, du$ *and* $\mathrm{Main}(t)$*, namely,*

$$\int_U \delta(t - u^{2k})\varphi(u)\, du - \mathrm{Main}(t)$$

can be bounded by a linear combination of terms of the form $t^{\lambda-1} \log^j(t^{-1})$ *for* $j = 0, \ldots, \theta - 2$ *and* $t^{\lambda-1+\mu/2} \log^{j'}(t^{-1})$ *for* $j' = 0, \ldots, d' - \theta - 1$*, where* $\mu = \mu_{\min}$ *or* $\mu = \mu_{\max}$*.*

(Proof)

By substituting

$$u_1 = t^{1/(2k_1)} u_2^{-k_2/k_1} \cdots u_d^{-k_{d'}/k_1},$$

we obtain

$$\frac{u^h}{2k_1 u_1^{2k_1-1} u_2^{2k_2} \cdots u_{d'}^{2k_{d'}}}$$

$$= \frac{1}{2k_1} t^{\frac{h_1 - 2k_1 + 1}{2k_1}} u_2^{-\frac{(h_1+1)k_2}{k_1} + h_2} \cdots u_{d'}^{-\frac{(h_1+1)k_{d'}}{k_1} + h_{d'}}$$

$$= \frac{1}{2k_1} t^{\lambda-1} u_2^{-1} \cdots u_\theta^{-1} u_{\theta+1}^{-2\lambda k_{\theta+1} + h_{\theta+1}} \cdots u_{d'}^{-2\lambda k_{d'} + h_{d'}}.$$

On the other hand, the set U_t is given by

$$U_t = \left\{ t^{\frac{1}{2k_2}} u_3^{-\frac{k_3}{k_2}} \cdots u_d^{-\frac{k_{d'}}{k_2}} < u_2 < 1, \right.$$
$$\left. t^{\frac{1}{2k_3}} u_4^{-\frac{k_4}{k_3}} \cdots u_d^{-\frac{k_{d'}}{k_3}} < u_3 < 1, \cdots, t^{\frac{1}{2k_{d'}}} < u_{d'} < 1 \right\}.$$

Integrating u_2^{-1} over the range

$$t^{\frac{1}{2k_2}} u_3^{-\frac{k_3}{k_2}} \cdots u_d^{-\frac{k_{d'}}{k_2}} < u_2 < 1,$$

we obtain

$$\int_{u_2} u_2^{-1} du_2 = \frac{1}{k_2} \log(t^{-\frac{1}{2}} u_3^{k_3} \cdots u_{d'}^{k_{d'}}).$$

Next, integrating

$$\frac{1}{k_2} \log(t^{-\frac{1}{2}} u_3^{k_3} \cdots u_{d'}^{k_{d'}}) u_3^{-1}$$

over u_3 in the range $t^{\frac{1}{2k_3}} u_4^{-\frac{k_4}{k_3}} \cdots u_d^{-\frac{k_{d'}}{k_2}} < u_3 < 1$, we obtain

$$\int_{u_2, u_3} u_2^{-1} u_3^{-1} du_2 du_3$$

$$= \frac{1}{2k_2k_3}\log^2(t^{-\frac{1}{2}}u_4^{k_4}\cdots u_{d'}^{k_{d'}}).$$

By repeating this process, integrating $u_2^{-1}\cdots u_\theta^{-1}$ over U_t yields

$$\frac{1}{(\theta-1)!k_2k_3\cdots k_\theta}\log^{\theta-1}(t^{-\frac{1}{2}}u_{\theta+1}^{k_{\theta+1}}\cdots u_{d'}^{k_{d'}}).$$

Next, in the range

$$t^{\frac{1}{2k_{\theta+1}}}u_{\theta+2}^{-\frac{k_{\theta+2}}{k_{\theta+1}}}\cdots u_{d'}^{-\frac{k_{d'}}{k_{\theta+1}}} < u_{\theta+1} < 1,$$

for $\mu > 0$, we integrate

$$u_{\theta+1}^{k_{\theta+1}\mu-1}\log^{\theta-1}(t^{-\frac{1}{2}}u_{\theta+1}^{k_{\theta+1}}\cdots u_{d'}^{k_{d'}})$$

with respect to $u_{\theta+1}$, obtaining

$$\int u_{\theta+1}^{k_{\theta+1}\mu-1}\log^{\theta-1}(t^{-\frac{1}{2}}u_{\theta+1}^{k_{\theta+1}}\cdots u_{d'}^{k_{d'}})du_{\theta+1} \quad (1)$$

$$= \sum_{j=1}^{\theta}\frac{(-k_{\theta+1})^{j-1}(\theta-1)!}{(k_{\theta+1}\mu)^j(\theta-j)!}\log^{\theta-j}(t^{-\frac{1}{2}}u_{\theta+2}^{k_{\theta+2}}\cdots u_{d'}^{k_{d'}})$$

$$-\frac{(-k_{\theta+1})^{\theta-1}(\theta-1)!}{(k_{\theta+1}\mu)^\theta}(t^{\frac{1}{2}}u_{\theta+2}^{-k_{\theta+2}}\cdots u_{d'}^{-k_{d'}})^\mu$$

$$= \sum_{j=1}^{\theta}\frac{(-1)^{j-1}(\theta-1)!}{k_{\theta+1}\mu^j(\theta-j)!}\log^{\theta-j}(t^{-\frac{1}{2}}u_{\theta+2}^{k_{\theta+2}}\cdots u_{d'}^{k_{d'}})$$

$$-\frac{(-1)^{\theta-1}(\theta-1)!}{k_{\theta+1}\mu^\theta}(t^{\frac{1}{2}}u_{\theta+2}^{-k_{\theta+2}}\cdots u_{d'}^{-k_{d'}})^\mu.$$

By assumption, we obtain $u_j^{-2k_j(\lambda-\frac{h_{d'}+1}{(2k_{d'})})-1}$ $\le u_j^{-2k_j(\lambda-\frac{(h_{j+1}+1)}{(2k_{j+1})})-1} \le u_j^{-2k_j(\lambda-\frac{(h_{\theta+1}+1)}{(2k_{\theta+1})})-1}$.

Additionally, evaluating the integral over U_t, we find that the integral is a linear combination of terms of the form $\log^j(t^{-1/2})$ and $t^{\mu/2}\log^{j'}(t^{-1/2})$, and by iterating equation (1), we obtain Main(t). (End of proof)

Theorem 4.3. *Let $t' = nt = nK(\pi(u)) = nu_1^{2k_1}\cdots u_d^{2k_d}$. We have $I =$*

$$= \int \exp(-nK(w))\varphi(w)\,dw$$

$$= \sum_u \int \exp(-nu_1^{2k_1}\cdots u_d^{2k_d})u_1^{h_1}\cdots u_d^{h_d}b(u)\,du$$

$$= \sum_u \int \exp(-nt)\int \delta(t-u_1^{2k_1}\cdots u_d^{2k_d})u_1^{h_1}\cdots u_d^{h_d}b(u)\,dt$$

$$= \sum_{u'} \int \exp(-nt)$$

$$\times\Big(c_{u'}^* t^{\lambda-1}\log^{\theta-1}(t^{-1}) + o(t^{\lambda-1}\log^{\theta-1})(t^{-1})\Big)\,dt$$

$$= \sum_{u'}\frac{c_{u'}^*\log^{\theta-1}(n)}{n^\lambda}\int \exp(-t')t'^{\lambda-1}dt'$$

$$+o(\frac{\log^{\theta-1}(n)}{n^\lambda}).$$

where u' is a local coordinate with λ and θ, and $c_{u'}^* = \frac{1}{(\theta-1)!2^\theta\prod_{j=1}^{d'}k_j\prod_{j=\theta+1}^{d'}(-2\lambda+\frac{h_j+1}{k_j})\prod_{j=d'+1}^{d}(h_j+1)}$.

Therefore, Theorem 4.3 leads to Theorem 2.2 without the use of the Mellin transform or zeta function.

5. Conclusion

In this paper, we examined the learning coefficients for deep neural networks with linear units and ReLU units. We demonstrated that the learning coefficients for ReLU units can be considered equivalent to those of neural networks with linear units by partitioning the input vector space. This result highlights the efficiency of ReLU units and extends a theoretical finding from (Aoyagi, 2024), which established that in networks with linear units, the learning coefficient decreases as the number of layers increases, revealing the double descent phenomenon. Our findings show that this property also holds for networks with ReLU units. Double descent is the phenomenon in which both generalization error and training error decrease (Nakkiran et al., 2020). Furthermore, in this paper, we provided an alternative proof for the order of the δ function without using the Mellin transformation or the zeta function. This proof method offers an intuitive understanding of Watanabe theory and is particularly useful because it does not require advanced mathematics. This allows for the clarification of not only the leading term but also additional terms, contributing to the further development of the theory. Recently, these theoretical values of the learning coefficients have been effectively utilized in numerical experiments, including information criteria, Markov chain Monte Carlo methods (Nagata & Watanabe, 2008a;b), and model selection techniques.

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Impact Statement

This study investigates the learning coefficients of deep neural networks. While previous research has analyzed three-layer neural networks, the first study on deep networks with linear units was presented in (Aoyagi, 2024). In this paper, we extend the analysis to nonlinear activation functions, specifically the ReLU function. The concept of learning coefficients originates from Bayesian machine learning and serves as a metric for evaluating model quality. Recent studies have shown that parameters with small learning coefficients tend to be highly stable during the learning process, making them valuable for theoretical research (Furman & Lau, 2024). While many studies focus on numerical experiments, theoretical research like ours not only provides a foundation for experimental findings but also offers valuable direction for future research. For instance, our demonstration that increasing the number of layers reduces the learning coefficient theoretically supports the efficiency of deep learning. Additionally, our analysis of the theoretical behavior of learning coefficients provides insights into the optimal number of layers, offering useful guidance for experimental and numerical research.

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